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**SJSU ID:** 020774013

**Objective.** This project prepares the full US Accidents dataset for a future four-class model of recorded accident `Severity`. The analysis deliberately separates (A) the raw imported data, (B) an analysis dataset after only objective, model-independent corrections, and (C) a future model-ready representation whose imputation, encoding, feature exclusions and learned transformations are specified only after EDA. The report therefore treats data quality, data collection, prediction-time availability and distribution shift as part of the modeling problem rather than as preprocessing details.

# 1 Dataset and Analytical Scope

The assignment uses the Kaggle US Accidents (2016–2023) dataset. The full `US_Accidents_March23.csv` file was loaded with an explicit PyArrow schema. The raw file contains **7,728,394 rows and 46 columns** and is approximately **3.06 GB**. **All four parts use the full file; the 500,000-row fallback sample was not used.** Recorded start times range from 14 January 2016 through 31 March 2023, cover 49 states, and come from three providers (`Source1`, `Source2`, and `Source3`).

The target, `Severity`, is complete and restricted to the four documented levels 1–4. It is retained as a four-class outcome rather than collapsed into a binary or ordinal surrogate. The raw schema combines provider metadata, timestamps, geography, weather measurements, road-context indicators, daylight indicators and free text (Table 1).

| Feature group | Examples and analytical role |
|---|---|
| Identifier and target | `ID`; `Severity` (prediction target) |
| Provider and time | `Source`, `Start_Time`, `End_Time`, `Weather_Timestamp` |
| Accident extent | `End_Lat`, `End_Lng`, `Distance(mi)` |
| Location | Start coordinates, street, city, county, state, ZIP code, airport code, timezone |
| Weather | Temperature, wind chill, humidity, pressure, visibility, wind direction/speed, precipitation, condition |
| Road context | Twelve Boolean infrastructure flags such as crossing, junction, railway, stop and traffic signal |
| Daylight | Sunrise/sunset plus civil, nautical and astronomical twilight |
| Free text | `Description` |

**Table 1:** Semantic groups in the raw schema.

A central characteristic of this dataset is that observations are not identically distributed across providers and time. Provider coverage changes over the seven-year window, the weather vocabulary changes abruptly, several missingness patterns are tied to collection period, and the recorded target distribution itself shifts within providers. The subsequent EDA therefore distinguishes statistical association from changes in how the data were collected or recorded.

# 2 Data Cleaning and Validation

Cleaning was intentionally conservative. Only corrections that are defensible without selecting a model family were applied. Missing-value imputation, encoding, scaling, clipping, winsorization and feature selection were not applied to the analysis dataset because those choices depend on the prediction design.

## 2.1 Objective cleaning actions

| Issue | Evidence | Action |
|---|---|---|
| Constant columns | `Country` is always US and `Turning_Loop` is always False | Drop both zero-variance columns. |
| Timestamp representation | Three time fields were text in up to three formats; all non-missing values parsed successfully | Convert `Start_Time`, `End_Time` and `Weather_Timestamp` to `datetime64`. |
| Hidden duplicate records | 102,338 copies appeared after excluding unique `ID`; timestamp normalization revealed 15,357 additional copies | Remove **117,695** confirmed later copies and keep the first occurrence. Retain 91,976 event-key overlaps that differ in other fields. |
| Whitespace | 1,696,520 `Street` values and 4,454 `Description` values had leading/trailing whitespace | Trim leading and trailing whitespace. |
| Wind direction coding | 24 raw labels represented 18 semantic states | Map full-word directions to 16-point abbreviations plus `CALM` and `VAR`; 1,081,609 rows are affected. |
| Placeholder text | 34 descriptions were exactly "Unknown" | Convert those placeholders to missing. |
| Definitionally invalid numeric value | Three `Pressure(in)` values were 0 although atmospheric pressure must be positive | Convert only those three readings to missing. |

**Table 2:** Objective corrections applied before EDA.

The timestamp audit also found no end time earlier than its start, but 4,229 records have durations above one year. That makes `End_Time` unreliable as a direct measure of the event and, more importantly, it is not necessarily known when an accident is first reported. Likewise, 93,903 weather timestamps are more than one hour from `Start_Time`. These observations are retained for EDA rather than silently removed.

## 2.2 Validation of the analysis dataset

After objective cleaning, the analysis dataset contains **7,610,699 rows and 44 columns**. Validation confirms that no proven duplicate remains, all three timestamp columns are parsed, and `Severity` contains no missing or invalid labels. The class counts are 65,570 / 6,048,391 / 1,295,235 / 201,503 for Severity 1–4, respectively.

| Check | Raw | After objective cleaning |
|---|---|---|
| Rows | 7,728,394 | 7,610,699 |
| Columns | 46 | 44 |
| Duplicate rows (`ID` excluded) | 117,695 | 0 |
| Timestamp parse failures | — | 0 |
| Timestamp columns stored as datetime | 0 of 3 | 3 of 3 |
| Target missing / invalid | 0 / 0 | 0 / 0 |
| Columns with missing values | 22 | 22 |
| Approx. in-memory size | 2.37 GB | 1.87 GB |

**Table 3:** Validation checkpoint before EDA.

Twenty-two columns still contain missing values after objective cleaning. Figure 1 shows the raw-data missingness profile from the initial audit; Table 3 confirms that the number of affected columns

remains 22 after cleaning. This is intentional. Missing end coordinates are provider-structural, while weather missingness changes sharply with collection period. Treating all gaps as random and globally imputing them before the train/test split would obscure the data-generation process and risk leakage.

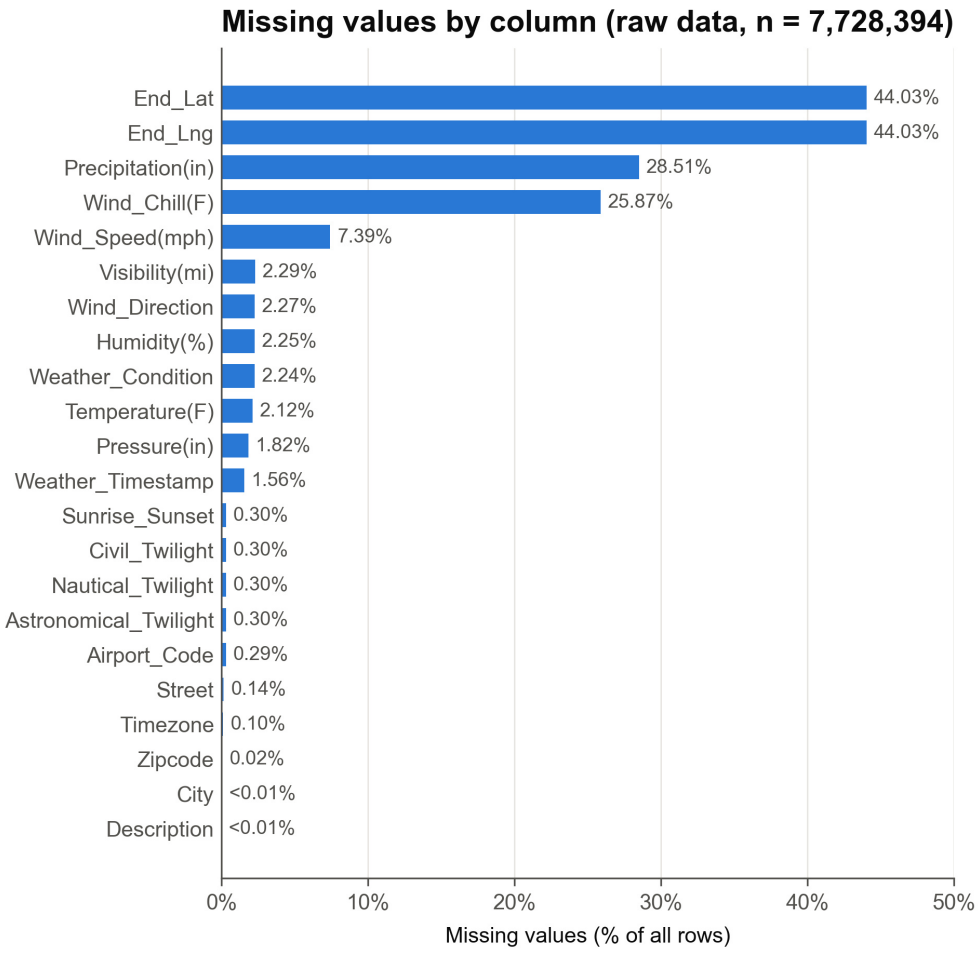



**Figure 1:** Missing-value rates in the raw dataset during the initial audit. Missingness is retained for analysis rather than treated as an automatic row-deletion criterion.

Extreme numeric values were handled by the same principle. The data contain values far from the bulk, including 207 °F temperature readings, 1,087 mph wind speed and 36.47 inches of precipitation, but the notebook uses the 1.5×IQR rule only to inspect distributions. The dataset does not provide documented validity bounds for these measurements, so a generic IQR rule would conflate statistical rarity with invalidity; it is especially unsuitable for zero-inflated variables such as precipitation and visibility. Therefore, only the three zero-pressure readings that violate a definitionally necessary constraint were changed at this stage; the other extremes are retained and flagged as modeling/validation concerns rather than silently deleted.

## 3 Exploratory Data Analysis

The EDA proceeds from univariate structure to collection effects, bivariate associations with `Severity`, and finally focused multivariate comparisons. The goal is not to interpret every pooled difference as a stable feature effect. Instead, the analysis asks whether the pattern survives comparison within similar providers, periods, locations and road contexts.

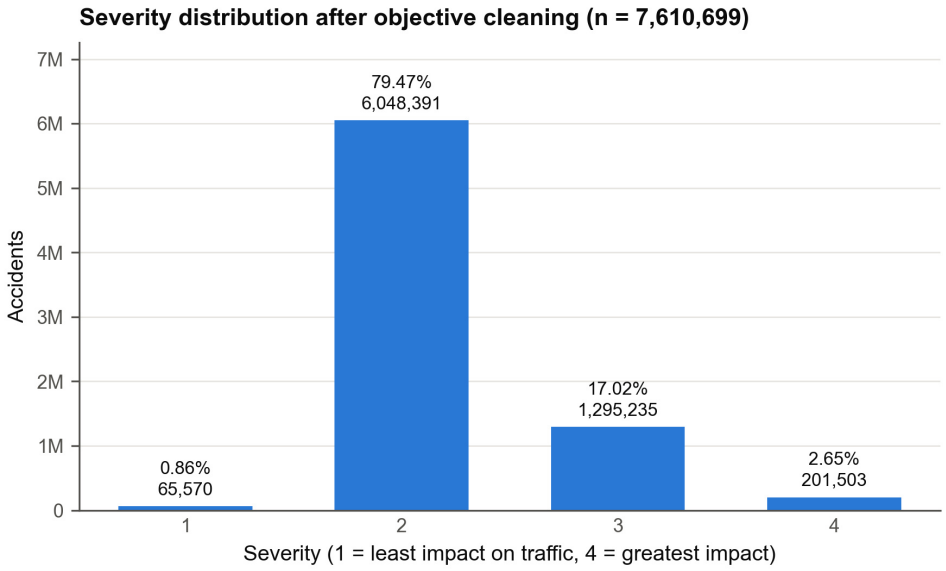


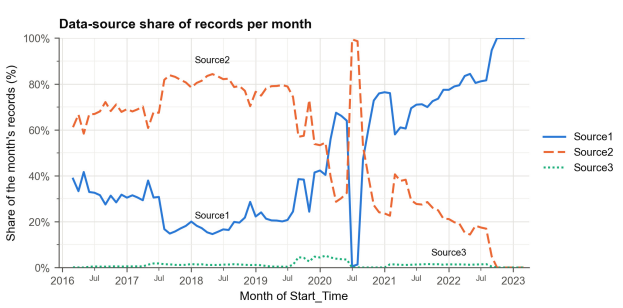

**Figure 2:** Distribution of the four Severity levels in the cleaned analysis dataset.

## 3.1 Target imbalance

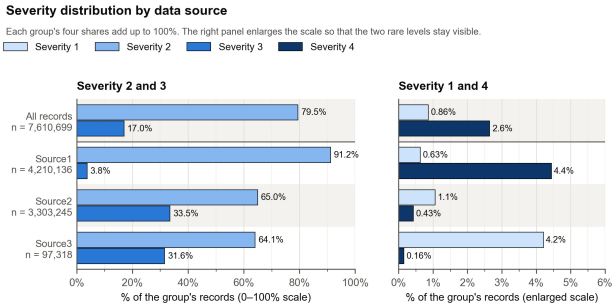
The target is highly imbalanced: **79.47%** of records are Severity 2, while only **0.86%** are Severity 1. The imbalance ratio between the largest and smallest class is 92.2. A classifier that always predicts Severity 2 would already achieve 79.47% accuracy, but only 25.00% balanced accuracy and macro-F1 of 0.221. For this reason, overall accuracy cannot be the primary evaluation metric; macro-F1, balanced accuracy and per-class recall are needed alongside it.

## 3.2 Collection process and source effects

Provider coverage changes substantially over time. Source1 spans January 2016–March 2023, whereas Source2 and Source3 end in September 2022. Source1 contributes only 0.5% and 1.3% of records in July and August 2020, respectively. Therefore, record volume cannot be read directly as accident frequency, and temporal patterns must be interpreted together with provider composition.




**(a)** Provider share by month.

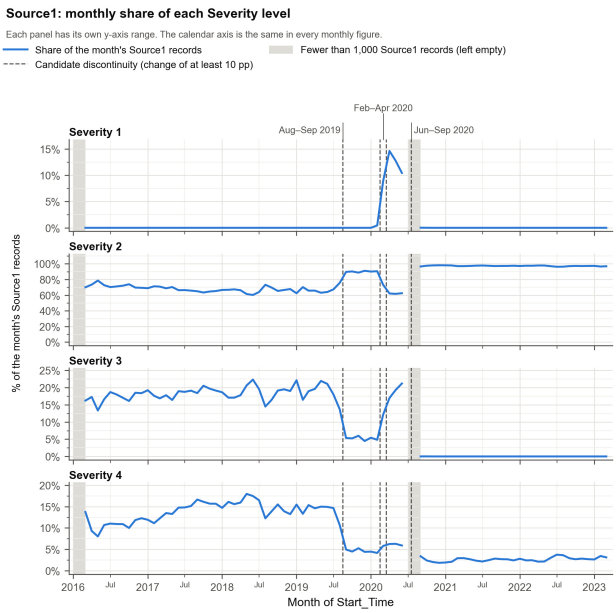



**(b)** Severity distribution by provider.

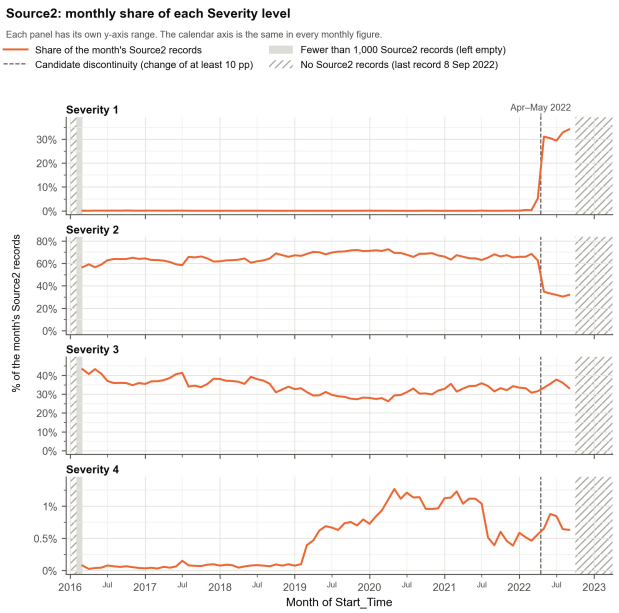
**Figure 3:** The provider mix changes over time, and the recorded Severity distribution differs substantially across providers.

The provider–Severity association is substantial: Cramér's $V$ is **0.29** over all records and ranges from 0.20 to 0.48 within individual years. The difference is not explained solely by which years each provider covers. Within-provider target distributions also change abruptly: Source1 records no Severity 3 from September 2020 onward, while Source2 and Source3 show a large increase in Severity 1 beginning in 2022.

The observable case mix does change around some of these boundaries, but it does not explain the




**(a)** Source1.




**(b)** Source2.

**Figure 4:** Monthly Severity composition within the two large providers. Abrupt changes remain visible even without mixing providers.

largest target shifts. Direct standardization using comparable state, calendar, road-class and daylight strata covers 93.5–99.5% of the records in the primary comparisons. For example, standardizing Source1 before its September 2020 boundary changes the reference distribution only modestly, whereas the observed comparison period still moves from 17.7% to 0% Severity 3. Likewise, Source2's April 2022 transition occurs with an almost unchanged standardized reference mix. Cross-provider differences also persist within shared State × Quarter strata and among 4,073 candidate same-accident pairs. These checks support a conclusion of *distribution shift*; they do not establish whether its cause is traffic conditions, coverage, provider selection, or coding/reporting practice.

Two additional collection artifacts reinforce that conclusion. First, the weather vocabulary changes in April 2019: labels such as "Clear" and "Overcast" disappear while "Fair" and "Cloudy" become common. Second, missing wind chill, precipitation and wind speed all drop sharply in the same month (for example, missing precipitation falls from 87.1% in March 2019 to 3.7% in April). Missingness and raw weather terminology therefore partly encode the collection regime.

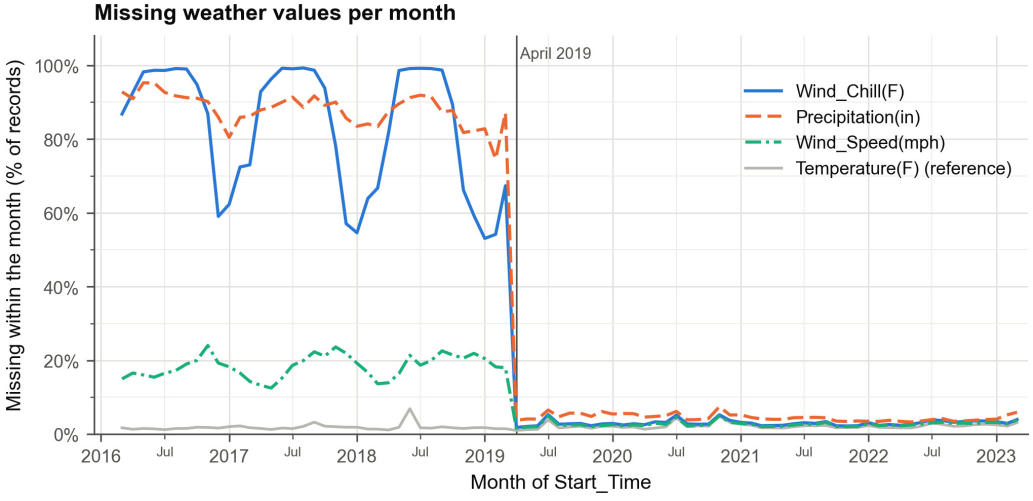



**Figure 5:** Weather-field missingness over time. The April 2019 break coincides with the weather-terminology change.

### 3.3 Bivariate relationships with Severity

The bivariate analysis examined time, weather, geography, road context, daylight, missingness and prediction-time leakage candidates. Several pooled patterns are large, but provider-stratified views show that magnitude and even direction can depend on how the records were collected.

**Time.** Severity 4 is most common near midnight (6.6% at 00:00 versus 1.7% at 07:00). Weekends show more Severity 4 overall, but the strongest weekday effect is the Source2-specific increase in Severity 3 on weekends (56.0% versus 30.9%). Year and month are strongly entangled with the source/period shifts: 97.4% of all Severity 1 records occur in March–June 2020 or April–September 2022.

**Weather.** Pooled snow/ice records contain more Severity 4, and rain/drizzle records contain more Severity 3, but the seven numeric weather measurements have weak monotonic relationships with Severity: $|\rho| \leq 0.090$ overall and within each large provider. Weather therefore needs contextual adjustment before its practical contribution can be judged.

**Geography.** Severity shares vary substantially among states and cities, but state rankings do not agree equally across providers. Across states observed by both large providers, the Spearman rank correlations of state-level shares are $-0.05$, 0.21, 0.47 and 0.50 for Severity 1–4. Geographic variables carry information, but part of a pooled state effect reflects the provider covering that state.

**Road and daylight context.** Traffic signals and crossings are associated with more Severity 1 and less Severity 3, while junctions and interstates are associated with more Severity 3. Night records contain more Severity 4 than day records. These relationships appear in both providers in several cases, but their strengths can differ sharply; for example, Source2 records 79.4% Severity 3 on interstates compared with 20.9% off interstates, while Source1 shows only 5.4% versus 3.1%.

**Missingness.** The pooled association between missing weather measurements and Severity largely disappears when the April 2019 collection break is respected. Missing wind chill, for example, corresponds to 30.1% versus 12.4% Severity 3 overall, but 31.9% versus 32.1% before April 2019 and 9.5% versus 10.8% afterward. This is a concrete example of a pooled relationship created mainly by collection-period composition.

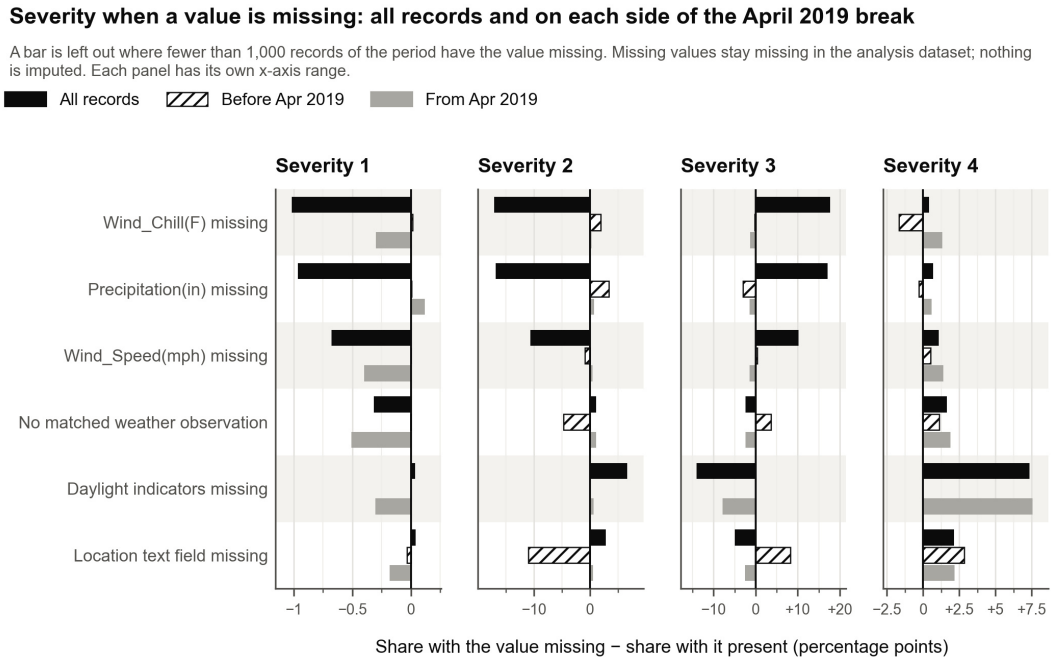



**Figure 6:** Difference in Severity shares when selected fields are missing, shown overall and on each side of the April 2019 collection break.

### 3.4 Focused multivariate EDA

The strongest bivariate relationships were re-estimated with direct standardization. This is not a predictive model: it compares the four-class Severity distribution of records with and without a feature after weighting comparable strata to the same distribution. Road features use Source × State × calendar-quarter strata; the temporal and weather analyses add appropriate road, geographic and seasonal context. The same contrasts are also estimated separately within Source1 and Source2.

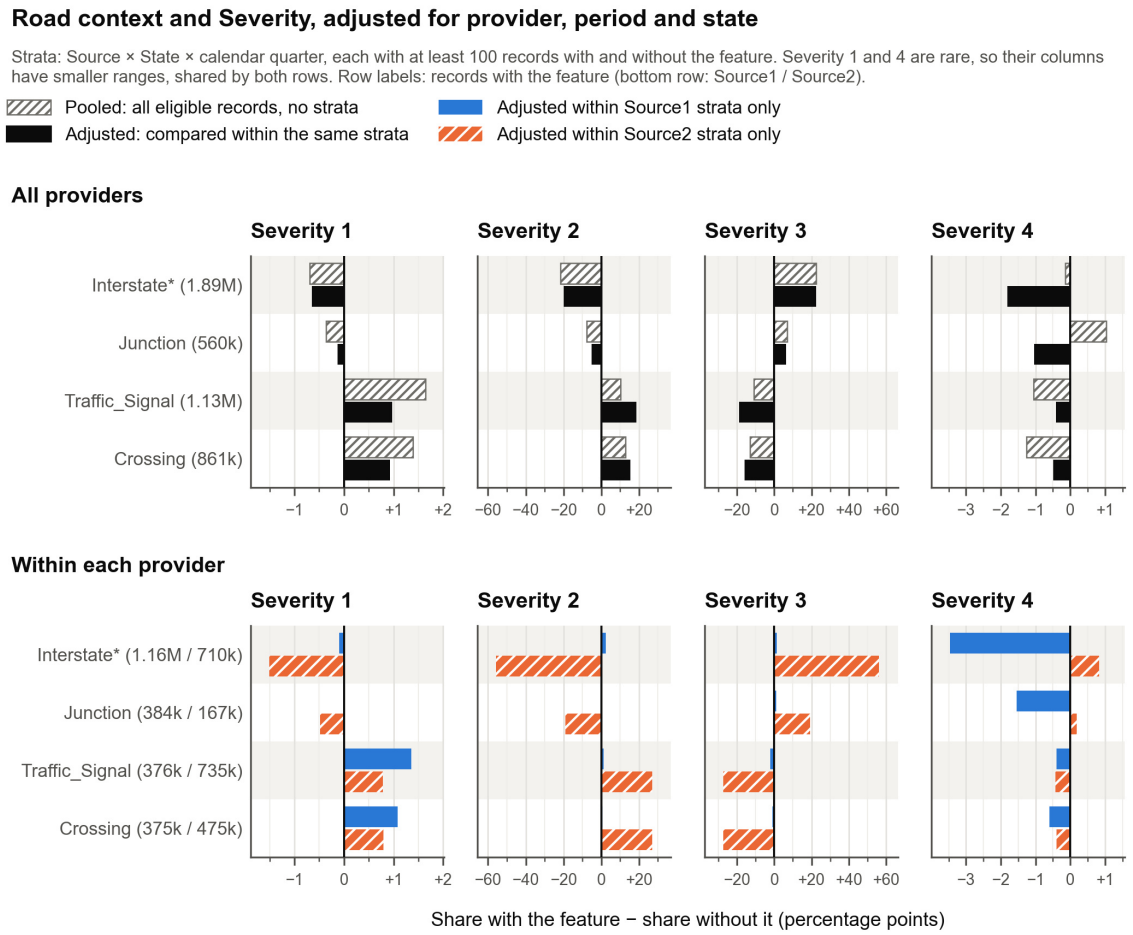



**Figure 7:** Pooled, adjusted and within-provider road-context differences. Values are percentage-point changes in each Severity share for records with versus without the feature.

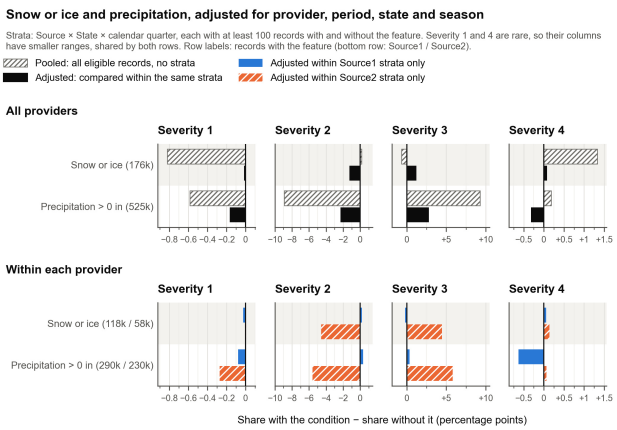
Road context retains the clearest adjusted associations. Traffic signals change from a pooled −10.9 point Severity 3 difference to −19.0 **points** after adjustment; crossings change from −13.0 to −15.9 **points**. Interstates remain at approximately **+22.5 points** Severity 3 and junctions at **+6.4 points**. The direction of these Severity 3 effects is stable, but most of their magnitude comes from Source2. Other levels can be source-specific: the adjusted interstate Severity 4 difference is −3.5 points within Source1 and +0.84 within Source2.

Nighttime Severity 4 is comparatively stable: its pooled +1.3 point difference becomes **+0.94 points** after adjustment and remains positive in both providers (+1.4 in Source1 and +0.22 in Source2). The weekend Severity 4 difference, by contrast, falls from +1.3 to +0.37 points; the adjusted weekend pattern is primarily a Source2 Severity 3 effect.

Weather effects attenuate more strongly. Snow or ice has a pooled +1.3 point Severity 4 difference but only **+0.08 points** after adjustment. Positive precipitation has a pooled +9.3 point Severity 3 difference that falls to **+2.8 points**; the adjusted difference is +0.37 in Source1 and +5.8 in Source2. The snow/ice analysis has good support among snow/ice records (86.9% lie in usable strata) but only 27.9% of all eligible records are in those common strata, so generalization beyond the overlap




**(a)** Temporal context.




**(b)** Weather context.

**Figure 8:** Directly standardized temporal and weather contrasts.

should remain cautious.

The largest binomial 95% half-width among all 32 adjusted contrasts is only 0.17 percentage points. Sampling variability is therefore small at this dataset scale; the main limitation is residual confounding and incomplete common support, not random error.

## 3.5 Prediction-time leakage boundary

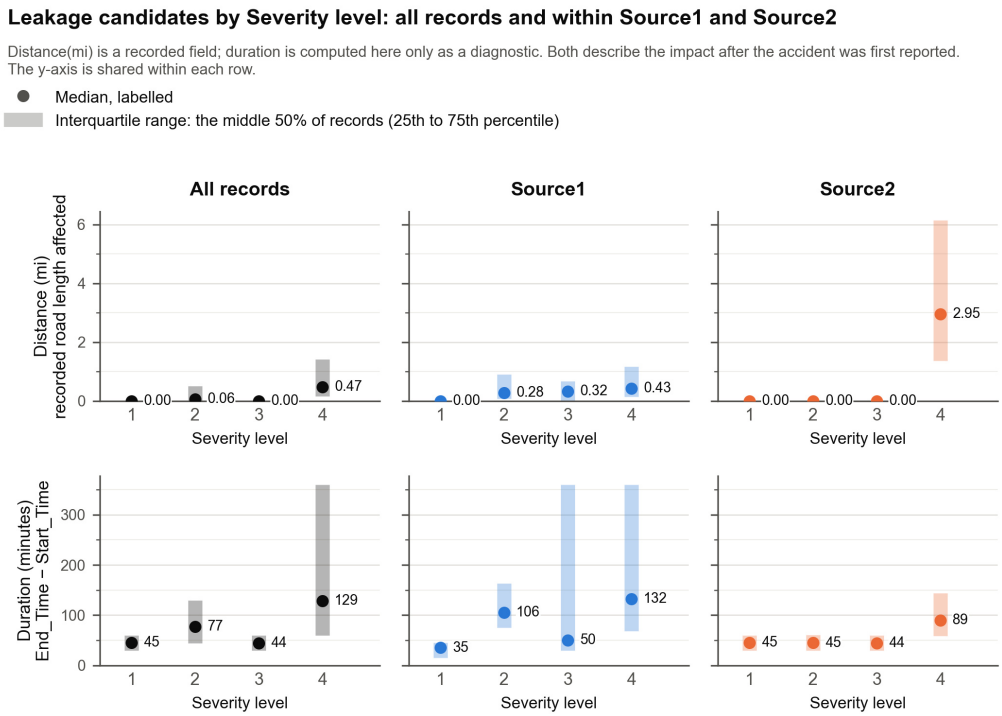



**Figure 9:** Accident extent and duration vary with Severity, but they are not appropriate baseline predictors when the prediction is made at initial report time.

`Distance(mi)` and duration are associated with Severity, but they describe realized impact after the accident is first reported. Their pooled Spearman correlations with Severity are $-0.135$ and $-0.176$, respectively, and both relationships change materially by provider. End-coordinate availability is even more provider-structural. These strong associations are therefore evidence of leakage risk rather than evidence for inclusion in a baseline model. `End_Time`, `End_Lat`, `End_Lng`, `Distance(mi)` and derived duration are excluded from the model-ready specification.

# 4 Key Insights and Modeling Implications

The EDA supports six conclusions that directly determine the preparation policy.

1. **Severity is severely imbalanced.** Severity 2 contains 79.47% of records and Severity 1 only 0.86%; class-sensitive metrics are required and any resampling or class weighting must occur after the split.
2. **Source and collection period shape the recorded target.** Large provider differences and abrupt within-provider changes persist after observable case-mix adjustment. Validation must therefore be chronological and results should also be reported by provider and collection period.
3. **Road context carries the clearest adjusted signal.** Traffic signals, crossings, interstates and junctions retain material associations, although several effects are much stronger in Source2 than Source1.
4. **Night is the most stable temporal contrast.** Its Severity 4 excess survives adjustment in both major providers, whereas several pooled hourly/weekend relationships are largely compositional or provider-specific.
5. **Weather contributes weaker, context-sensitive relationships and non-random missingness.** Raw weather labels partly encode an April 2019 vocabulary change, weather gaps follow the collection process, and pooled weather contrasts shrink after adjustment.
6. **Prediction-time availability overrides association strength.** End coordinates, distance and duration are strong correlates of recorded Severity but are excluded because they describe post-event impact rather than information available at the intended prediction time.

These findings imply a leakage-safe workflow: preserve valid observations and incomplete records, fit learned preprocessing only on the training partition, encode high-cardinality variables without using the target, harmonize changing weather vocabularies, and use a chronological test period to measure robustness under distribution shift.

# 5 Decision Table

Table 4 translates Parts 1 and 2 into a reproducible specification for the model-ready representation. Every data-derived operation—imputation medians, category frequencies, one-hot vocabularies, scaling, class weights or resampling—must be learned from the training partition only and then applied unchanged to validation and test data.

**Table 4:** Final data-preparation decision table.

| Column / issue | Decision | Because |
|---|---|---|
| `Severity` (target) | Keep all four levels (1–4); never impute, merge or relabel. Do not resample before splitting. Apply any class weighting or resampling to the training partition only. Evaluate with macro-F1, balanced accuracy and per-class recall in addition to accuracy. | The target is complete and valid, but Severity 2 holds 79.47% of records; a majority-only classifier would already reach 79.47% accuracy but only 25.00% balanced accuracy and 0.221 macro-F1 (Fig. 2). |
| `ID`, `Country`, `Turning_Loop` | Exclude from predictors. Keep `ID` only as a row key; drop `Country` and `Turning_Loop`. | `ID` is unique for all 7,728,394 raw rows; `Country` is 100% US and `Turning_Loop` is 100% False. |

*Continued on next page*

*Table 4 continued*

| Column / issue | Decision | Because |
|---|---|---|
| Duplicate and repeated records | Before splitting, remove rows identical in all non-`ID` fields after timestamp parsing, keeping the first. Keep unconfirmed event-key overlaps and cross-source candidate pairs; if reliable event grouping becomes available, split by event group. | 117,695 rows are proven copies, whereas other candidate repeats cannot be confirmed as the same accident. |
| `Start_Time` | Derive `Hour` (0–23) and `Is_weekend`. One-hot encode `Hour` with a training-only 24-level vocabulary and encode `Is_weekend` as 0/1. Use the timestamp to define the chronological holdout. Do not use raw timestamp, year or month as baseline predictors. | Hour/weekend contain predictive structure, whereas year/month strongly track collection regimes; 97.4% of all Severity 1 records occur in only two short periods (March–June 2020 or April–September 2022). |
| `End_Time`, `End_Lat`, `End_Lng`, `Distance(mi)` | Exclude from predictors, including any duration derived from `End_Time`; do not impute fields that will not be modeled. | They describe realized extent or duration and create prediction-time leakage risk (Fig. 9). |
| `Source` | Keep for provenance and subgroup reporting, but exclude from the baseline predictors. | Provider and Severity have Cramér's $V = 0.29$ overall and 0.20–0.48 within individual years; the cause is unresolved and may reflect coverage, selection or coding/reporting practice rather than accident conditions (Figs. 3–4). |
| `Description` | Exclude from the baseline tabular model. Treat a separate text model as future work requiring its own leakage review. | The raw file contains 3,761,578 distinct `Description` strings, making direct tabular encoding impractical, and provider-written text requires a separate prediction-time leakage review before use. |
| `Start_Lat`, `Start_Lng`, `State`, `Timezone` | Keep coordinates numeric. One-hot encode `State` and `Timezone` with training-only vocabularies; include a Missing level for `Timezone` and map unseen categories to all-zero one-hot vectors. | `State` has 49 levels and `Timezone` four; Severity shares vary materially by geography, while their cardinality remains small enough for one-hot representation. |
| `City`, `County`, `Zipcode`, `Airport_Code` | Merge letter-case variants; reduce ZIP+4 to the first five digits; frequency-encode each level using its share of training records. Map unseen or missing levels to 0. Do not target-encode. | The raw audit finds 13,678 cities, 1,871 counties, 825,094 ZIP-code strings and 2,045 airport codes; naive one-hot encoding is inefficient, while target encoding would use `Severity` directly. |
| `Street` | Exclude the raw name. Derive `Road_class` (Interstate, other named highway, other/unnamed street) using the notebook rule and one-hot encode it. | The raw audit contains 336,306 distinct street names, while the text-derived road-class proxy retains strong adjusted association with Severity (Fig. 7). |

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*Table 4 continued*

| Column / issue | Decision | Because |
|---|---|---|
| Numeric weather fields and `Weather_Timestamp` | Keep temperature, wind chill, humidity, pressure, visibility, wind speed and precipitation. Impute missing values with training-set medians. Add missing indicators for wind chill, wind speed and precipitation plus `No_weather_observation`. Drop the raw weather timestamp; keep rows whose weather observation is more than one hour from start time. Never treat missing precipitation as observed zero. | Weather gaps follow provider/period rather than chance: for example, missing precipitation falls from 87.1% in March 2019 to 3.7% in April 2019. Most observed precipitation values are genuine zeros (Figs. 5 and 6). |
| `Weather_Condition`, `Wind_Direction` | Map weather condition to the seven harmonized groups used in the EDA and wind direction to its 18 canonical states; one-hot encode both with a Missing level. | The raw audit contains 144 weather-condition labels and 24 wind-direction labels; the weather vocabulary changes abruptly in April 2019, and wind directions reduce to 18 canonical states after harmonization. |
| Twelve Boolean road flags | Keep all twelve as 0/1 predictors, including rare flags. | The columns are complete, and several retain material adjusted relationships with Severity: the adjusted Severity 3 difference is about +22.5 points on interstates, -19.0 at traffic signals, -15.9 at crossings and +6.4 at junctions (Fig. 7). |
| Daylight / twilight fields | Keep `Sunrise_Sunset` as Day/Night/Missing; exclude `Civil_Twilight`, `Nautical_Twilight` and `Astronomical_Twilight`. | The four indicators encode overlapping daylight definitions. Using `Sunrise_Sunset`, the adjusted nighttime Severity 4 difference is +0.94 percentage points and stays positive within both large providers (Fig. 8). |
| Missing values, extremes and learned preprocessing | Do not delete a record for a missing predictor. Retain extreme values unless a documented validity rule is violated; do not apply automatic IQR deletion, clipping or winsorization. Fit every data-derived preprocessing step on training data only. | Twenty-two columns remain incomplete after objective cleaning, and missingness changes sharply by collection period (Figs. 1 and 5). Broad IQR rules would treat rarity as invalidity, while fitting preprocessing on all data would leak validation/test information. |
| Evaluation design | Train on records before 1 April 2022 and use 1 April 2022–31 March 2023 as the primary chronological test period. Tune within the training period using a Severity-stratified random split. Report metrics overall, by `Source`, and by collection period (before Sep 2020, Sep 2020–Mar 2022, from Apr 2022). | Abrupt within-provider shifts make a random split alone overly optimistic about robustness (Figs. 3–4). |

# 6 RAPIDS Experiment

Part 4 repeats six core EDA task families from Part 2 on the full **7,610,699 × 44** analysis dataset derived from the full 7,728,394-row file; no 500k sample is used. Together these tasks cover the Part 2 analysis types: univariate profiling, collection/time effects, Severity groupings, weather/correlation analysis, road/string analysis, and focused multivariate standardization. The final run used a **Tesla**

**T4 (15.6 GB)**, **cuDF 26.02.01** and **CuPy 14.0.1**. Each analytical task is implemented once and called with either the pandas/NumPy objects or the cuDF/CuPy equivalents so that the analytical definitions remain unchanged.

### 6.1 Equivalent CPU and GPU operations

| EDA task | pandas / NumPy | cuDF / CuPy |
|---|---|---|
| Univariate profile | `value_counts`, `isna().sum`, `nunique`, `describe` | Corresponding cuDF methods |
| Collection effects | datetime accessors, `groupby(...).size()/mean()` | cuDF datetime accessors and `groupby` |
| Severity by source and time | pandas group-by counting | cuDF group-by counting |
| Weather groups and Spearman correlations | NumPy `bincount`, pandas `rank`, NumPy dot products | CuPy `bincount`, cuDF `rank`, CuPy dot products |
| Road class and road-feature contrasts | pandas string/regex methods, NumPy counting | cuDF GPU string/regex methods, CuPy counting |
| Direct standardization | NumPy `unique`, `bincount`, matrix products | CuPy `unique`, `bincount`, matrix products |

**Table 5:** Representative EDA operations reproduced with RAPIDS.

The full analysis table is copied to GPU memory once through Apache Arrow. The copy takes approximately **1.83 s** and occupies **1.71 GB** of GPU memory. For each task and each backend, the notebook performs one untimed warm-up followed by three timed runs and reports the median. GPU timing includes an explicit device synchronization because CUDA execution is asynchronous. Disk loading is excluded, and each timed GPU task includes transfer of its small aggregated result back to CPU memory.

### 6.2 Correctness checks

Performance is interpreted only after checking result equivalence. Integer counts must agree exactly. Floating-point outputs use a relative tolerance of $10^{-9}$, except for the numeric summary of 32-bit weather measurements, which uses $10^{-5}$. In addition, one key GPU result from each task is checked against the corresponding Part 2 object. All six tasks pass (Table 6).

| Task | Results | Largest diff. | Part 2 anchor checked on GPU | Status |
|---|---|---|---|---|
| Univariate profile | 4 | $4.7 \times 10^{-7}$ | Severity counts | Pass |
| Collection effects | 2 | 0 | Records per month and source | Pass |
| Severity by source and time | 4 | 0 | Cramér's $V$ between Source and Severity | Pass |
| Weather groups and correlations | 2 | $4.0 \times 10^{-14}$ | Spearman correlations with Severity | Pass |
| Road class and road features | 5 | 0 | Records per road class | Pass |
| Direct standardization | 3 | $3.7 \times 10^{-15}$ | Adjusted traffic-signal contrast and usable strata | Pass |

**Table 6:** CPU–GPU equivalence checks from the final notebook execution.

### 6.3 Runtime comparison

The compute-only workflow is **8.5× faster** on the T4 for this benchmark, and **5.5× faster** after charging the one-time data copy. The largest task-level speedup is **20.9×** for Severity-by-source/time groupings, while the road-class/string task is only **1.5×** faster. This variation is important: RAPIDS

| Task | pandas/NumPy (s) | cuDF/CuPy (s) | Speedup |
|---|---|---|---|
| Univariate profile | 9.23 | 0.682 | 13.5× |
| Collection effects | 1.63 | 0.099 | 16.4× |
| Severity by source and time | 2.54 | 0.122 | 20.9× |
| Weather groups and correlations | 11.59 | 0.650 | 17.8× |
| Road class and road features | 2.77 | 1.793 | 1.5× |
| Direct standardization | 1.45 | 0.098 | 14.8× |
| All six tasks | 29.21 | 3.445 | 8.5× |
| All six tasks, including one-time GPU copy | 29.21 | 5.277 | 5.5× |

**Table 7:** Median runtime after one warm-up and three timed runs. Disk loading is excluded on both sides; the final row includes the one-time host-to-device copy.

is not a constant-factor replacement for pandas. Vectorized counting, group-by operations and array arithmetic benefit strongly at this scale, whereas regular-expression/string processing carries more overhead.

The workflow also differs operationally. CPU and GPU memory are separate, so large transfers should occur once rather than inside every task. cuDF and CuPy are close to pandas and NumPy but not identical: missing categorical codes, group-by ordering and host/device conversion require explicit handling. Finally, valid GPU benchmarking requires warm-up and synchronization; otherwise initialization costs or asynchronous execution can make timing comparisons misleading.

# 7 Conclusion

The analysis shows that predicting recorded accident `Severity` is not only a feature-engineering problem but also a data-generation and validation problem. The cleaned dataset remains very large and information-rich, but the target is heavily imbalanced and its distribution changes sharply across providers and collection periods. Several intuitive pooled relationships—especially weather missingness and parts of the temporal/weather structure—weaken after contextual adjustment, while road context and nighttime Severity 4 remain more stable. At the same time, some of the strongest raw correlates, such as distance and duration, are inappropriate because they describe post-event impact.

These findings lead to a model-ready specification that preserves valid observations and incomplete records, excludes leakage, harmonizes changing representations, uses training-only preprocessing, handles high-cardinality variables without target leakage, and evaluates robustness with a chronological holdout plus provider/period subgroup reporting. The RAPIDS replication confirms that the same EDA definitions can be reproduced on GPU with numerically equivalent results and substantial acceleration for most large aggregation workloads, while also showing that GPU benefits depend on the operation rather than applying uniformly.